Conditional probability and Bayes' theorem

ST2334 · Lecture 3

Demo course material, written for OpenPapr

Conditional probability

- $P(A \mid B) = P(A \text{ and } B) / P(B)$, for $P(B) > 0$
- The probability of A once we know B has happened
- B becomes the new sample space

Example: two dice

- B: the sum is 8, so (2,6), (3,5), (4,4), (5,3) or (6,2)
- A: both dice show the same number
- P(A | B) = 1/5, while P(A) = 6/36 = 1/6

Multiplication rule

- P(A and B) = P(A | B) P(B) = P(B | A) P(A)
- Chain rule: P(A1 A2 A3) = P(A1) P(A2 | A1) P(A3 | A1 A2)

Law of total probability

- If B1, ..., Bk partition the sample space:
- P(A) = P(A | B1) P(B1) + ... + P(A | Bk) P(Bk)
- Split a hard probability into easy cases

Bayes' theorem

- $P(B \mid A) = P(A \mid B)\, P(B) / P(A)$
- Turns P(evidence | cause) into P(cause | evidence)
- Expand P(A) with the law of total probability

Example: a medical test

- 1% of people have the disease
- The test is positive for 95% of those who have it, and for 5% of those who do not
- P(positive) = 0.95(0.01) + 0.05(0.99) = 0.059
- P(disease | positive) = 0.0095 / 0.059 = 0.161, about 16%

The base-rate fallacy

- A positive result from an accurate test can still leave the disease unlikely
- When the condition is rare, false positives outnumber true positives

Independence

- A and B are independent if P(A and B) = P(A) P(B)
- Equivalently P(A | B) = P(A), when P(B) > 0
- Independent is not the same as mutually exclusive

Summary

- Conditioning: restrict to B and rescale
- Total probability: split into cases
- Bayes: reverse the conditioning